

500 Easy DSA Coding Practice Questions

Structured beginner-level problem bank for coding-platform ingestion. Each problem includes statement, constraints, I/O format, examples, visible tests, and hidden-test design guidance.

1. Find the maximum element in an array

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

2. Find the minimum element in an array

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 1 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

3. Calculate the sum of array elements

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 15 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

4. Reverse an array

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

5. Count even numbers in an array

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

6. Move all zeros to the end

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

7. Remove duplicates from a sorted array

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

8. Find the second largest element

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

9. Rotate an array left by one position

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

10. Check whether an array is sorted

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

11. Find the maximum element in an array — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

12. Find the minimum element in an array — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

13. Calculate the sum of array elements — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

14. Reverse an array — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

15. Count even numbers in an array — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

16. Move all zeros to the end — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

17. Remove duplicates from a sorted array — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

18. Find the second largest element — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

19. Rotate an array left by one position — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

20. Check whether an array is sorted — Variation 2

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

21. Find the maximum element in an array — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

22. Find the minimum element in an array — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

23. Calculate the sum of array elements — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

24. Reverse an array — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

25. Count even numbers in an array — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

26. Move all zeros to the end — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

27. Remove duplicates from a sorted array — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

28. Find the second largest element — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

29. Rotate an array left by one position — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

30. Check whether an array is sorted — Variation 3

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

31. Find the maximum element in an array — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

32. Find the minimum element in an array — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 1 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

33. Calculate the sum of array elements — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 15 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

34. Reverse an array — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

35. Count even numbers in an array — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

36. Move all zeros to the end — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

37. Remove duplicates from a sorted array — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

38. Find the second largest element — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

39. Rotate an array left by one position — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

40. Check whether an array is sorted — Variation 4

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

41. Find the maximum element in an array — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

42. Find the minimum element in an array — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

43. Calculate the sum of array elements — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

44. Reverse an array — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

45. Count even numbers in an array — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

46. Move all zeros to the end — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

47. Remove duplicates from a sorted array — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

48. Find the second largest element — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

49. Rotate an array left by one position — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

50. Check whether an array is sorted — Variation 5

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

51. Find the maximum element in an array — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the maximum element in an array — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

52. Find the minimum element in an array — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the minimum element in an array — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

53. Calculate the sum of array elements — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: calculate the sum of array elements — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

54. Reverse an array — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: reverse an array — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

55. Count even numbers in an array — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: count even numbers in an array — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

56. Move all zeros to the end — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: move all zeros to the end — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

57. Remove duplicates from a sorted array — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: remove duplicates from a sorted array — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

58. Find the second largest element — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: find the second largest element — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

59. Rotate an array left by one position — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: rotate an array left by one position — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

60. Check whether an array is sorted — Variation 6

Category: Arrays **Difficulty:** Easy **Tags:** arrays, beginner

Problem Statement: Solve the following task: check whether an array is sorted — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

61. Reverse a string

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: reverse a string. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

62. Check whether a string is a palindrome

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether a string is a palindrome. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

63. Count vowels in a string

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count vowels in a string. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

64. Count words in a sentence

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count words in a sentence. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

65. Convert lowercase letters to uppercase

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: convert lowercase letters to uppercase. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

66. Remove spaces from a string

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: remove spaces from a string. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

67. Find the first repeated character

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: find the first repeated character. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

68. Count frequency of each character

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count frequency of each character. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

69. Check whether two strings are anagrams

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether two strings are anagrams. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

70. Replace every space with a hyphen

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: replace every space with a hyphen. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

71. Reverse a string — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: reverse a string — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

72. Check whether a string is a palindrome — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether a string is a palindrome — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

73. Count vowels in a string — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count vowels in a string — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

74. Count words in a sentence — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count words in a sentence — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

75. Convert lowercase letters to uppercase — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: convert lowercase letters to uppercase — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

76. Remove spaces from a string — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: remove spaces from a string — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

77. Find the first repeated character — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: find the first repeated character — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

78. Count frequency of each character — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count frequency of each character — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

79. Check whether two strings are anagrams — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether two strings are anagrams — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

80. Replace every space with a hyphen — Variation 2

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: replace every space with a hyphen — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

81. Reverse a string — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: reverse a string — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

82. Check whether a string is a palindrome — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether a string is a palindrome — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

83. Count vowels in a string — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count vowels in a string — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

84. Count words in a sentence — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count words in a sentence — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

85. Convert lowercase letters to uppercase — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: convert lowercase letters to uppercase — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

86. Remove spaces from a string — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: remove spaces from a string — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

87. Find the first repeated character — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: find the first repeated character — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

88. Count frequency of each character — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count frequency of each character — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

89. Check whether two strings are anagrams — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether two strings are anagrams — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

90. Replace every space with a hyphen — Variation 3

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: replace every space with a hyphen — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

91. Reverse a string — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: reverse a string — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

92. Check whether a string is a palindrome — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether a string is a palindrome — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

93. Count vowels in a string — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count vowels in a string — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

94. Count words in a sentence — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count words in a sentence — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

95. Convert lowercase letters to uppercase — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: convert lowercase letters to uppercase — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

96. Remove spaces from a string — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: remove spaces from a string — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

97. Find the first repeated character — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: find the first repeated character — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

98. Count frequency of each character — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count frequency of each character — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

99. Check whether two strings are anagrams — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether two strings are anagrams — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

100. Replace every space with a hyphen — Variation 4

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: replace every space with a hyphen — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

101. Reverse a string — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: reverse a string — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

102. Check whether a string is a palindrome — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether a string is a palindrome — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

103. Count vowels in a string — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count vowels in a string — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

104. Count words in a sentence — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count words in a sentence — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

105. Convert lowercase letters to uppercase — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: convert lowercase letters to uppercase — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

106. Remove spaces from a string — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: remove spaces from a string — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

107. Find the first repeated character — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: find the first repeated character — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

108. Count frequency of each character — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: count frequency of each character — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

109. Check whether two strings are anagrams — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: check whether two strings are anagrams — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

110. Replace every space with a hyphen — Variation 5

Category: Strings **Difficulty:** Easy **Tags:** strings, beginner

Problem Statement: Solve the following task: replace every space with a hyphen — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

111. Implement linear search

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: implement linear search. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

112. Find the first occurrence of a target

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the first occurrence of a target. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

113. Find the last occurrence of a target

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the last occurrence of a target. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

114. Binary search in a sorted array

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: binary search in a sorted array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

115. Count occurrences of a target

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: count occurrences of a target. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

116. Find insertion position in a sorted array

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find insertion position in a sorted array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

117. Implement linear search — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: implement linear search — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

118. Find the first occurrence of a target — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the first occurrence of a target — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

119. Find the last occurrence of a target — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the last occurrence of a target — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

120. Binary search in a sorted array — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: binary search in a sorted array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

121. Count occurrences of a target — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: count occurrences of a target — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

122. Find insertion position in a sorted array — Variation 2

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find insertion position in a sorted array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

123. Implement linear search — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: implement linear search — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

124. Find the first occurrence of a target — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the first occurrence of a target — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

125. Find the last occurrence of a target — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the last occurrence of a target — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

126. Binary search in a sorted array — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: binary search in a sorted array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

127. Count occurrences of a target — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: count occurrences of a target — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

128. Find insertion position in a sorted array — Variation 3

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find insertion position in a sorted array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

129. Implement linear search — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: implement linear search — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

130. Find the first occurrence of a target — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the first occurrence of a target — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

131. Find the last occurrence of a target — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the last occurrence of a target — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

132. Binary search in a sorted array — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: binary search in a sorted array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

133. Count occurrences of a target — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: count occurrences of a target — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

134. Find insertion position in a sorted array — Variation 4

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find insertion position in a sorted array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

135. Implement linear search — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: implement linear search — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

136. Find the first occurrence of a target — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the first occurrence of a target — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

137. Find the last occurrence of a target — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find the last occurrence of a target — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

138. Binary search in a sorted array — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: binary search in a sorted array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

139. Count occurrences of a target — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: count occurrences of a target — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

140. Find insertion position in a sorted array — Variation 5

Category: Searching **Difficulty:** Easy **Tags:** searching, beginner

Problem Statement: Solve the following task: find insertion position in a sorted array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

141. Implement bubble sort

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement bubble sort. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

142. Implement selection sort

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement selection sort. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

143. Implement insertion sort

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement insertion sort. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

144. Sort an array of 0s and 1s

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort an array of 0s and 1s. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

145. Merge two sorted arrays

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: merge two sorted arrays. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

146. Sort strings lexicographically

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort strings lexicographically. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

147. Implement bubble sort — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement bubble sort — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

148. Implement selection sort — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement selection sort — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

149. Implement insertion sort — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement insertion sort — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

150. Sort an array of 0s and 1s — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort an array of 0s and 1s — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

151. Merge two sorted arrays — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: merge two sorted arrays — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

152. Sort strings lexicographically — Variation 2

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort strings lexicographically — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

153. Implement bubble sort — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement bubble sort — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

154. Implement selection sort — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement selection sort — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

155. Implement insertion sort — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement insertion sort — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

156. Sort an array of 0s and 1s — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort an array of 0s and 1s — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

157. Merge two sorted arrays — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: merge two sorted arrays — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

158. Sort strings lexicographically — Variation 3

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort strings lexicographically — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

159. Implement bubble sort — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement bubble sort — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

160. Implement selection sort — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement selection sort — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

161. Implement insertion sort — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement insertion sort — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

162. Sort an array of 0s and 1s — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort an array of 0s and 1s — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

163. Merge two sorted arrays — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: merge two sorted arrays — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

164. Sort strings lexicographically — Variation 4

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort strings lexicographically — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

165. Implement bubble sort — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement bubble sort — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

166. Implement selection sort — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement selection sort — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

167. Implement insertion sort — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: implement insertion sort — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

168. Sort an array of 0s and 1s — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort an array of 0s and 1s — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

169. Merge two sorted arrays — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: merge two sorted arrays — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

170. Sort strings lexicographically — Variation 5

Category: Sorting **Difficulty:** Easy **Tags:** sorting, beginner

Problem Statement: Solve the following task: sort strings lexicographically — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

171. Find duplicates using a hash set

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find duplicates using a hash set. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

172. Count element frequencies

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: count element frequencies. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

173. Find the unique element

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the unique element. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

174. Check whether two arrays have common elements

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: check whether two arrays have common elements. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

175. Find the first non-repeating character

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the first non-repeating character. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

176. Find duplicates using a hash set — Variation 2

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find duplicates using a hash set — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

177. Count element frequencies — Variation 2

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: count element frequencies — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

178. Find the unique element — Variation 2

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the unique element — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

179. Check whether two arrays have common elements — Variation 2

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: check whether two arrays have common elements — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

180. Find the first non-repeating character — Variation 2

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the first non-repeating character — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

181. Find duplicates using a hash set — Variation 3

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find duplicates using a hash set — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

182. Count element frequencies — Variation 3

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: count element frequencies — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

183. Find the unique element — Variation 3

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the unique element — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

184. Check whether two arrays have common elements — Variation 3

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: check whether two arrays have common elements — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

185. Find the first non-repeating character — Variation 3

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the first non-repeating character — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

186. Find duplicates using a hash set — Variation 4

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find duplicates using a hash set — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

187. Count element frequencies — Variation 4

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: count element frequencies — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

188. Find the unique element — Variation 4

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the unique element — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

189. Check whether two arrays have common elements — Variation 4

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: check whether two arrays have common elements — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

190. Find the first non-repeating character — Variation 4

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the first non-repeating character — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

191. Find duplicates using a hash set — Variation 5

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find duplicates using a hash set — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

192. Count element frequencies — Variation 5

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: count element frequencies — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

193. Find the unique element — Variation 5

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the unique element — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

194. Check whether two arrays have common elements — Variation 5

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: check whether two arrays have common elements — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

195. Find the first non-repeating character — Variation 5

Category: Hashing **Difficulty:** Easy **Tags:** hashing, beginner

Problem Statement: Solve the following task: find the first non-repeating character — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

196. Find a pair with a given sum

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find a pair with a given sum. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

197. Remove duplicates using two pointers

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: remove duplicates using two pointers. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

198. Reverse vowels in a string

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: reverse vowels in a string. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

199. Find maximum sum subarray of size K

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find maximum sum subarray of size k. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

200. Find the longest substring without repeating characters

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find the longest substring without repeating characters. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

201. Find a pair with a given sum — Variation 2

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find a pair with a given sum — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

202. Remove duplicates using two pointers — Variation 2

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: remove duplicates using two pointers — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

203. Reverse vowels in a string — Variation 2

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: reverse vowels in a string — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

204. Find maximum sum subarray of size K — Variation 2

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find maximum sum subarray of size k — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

205. Find the longest substring without repeating characters — Variation 2

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find the longest substring without repeating characters — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

206. Find a pair with a given sum — Variation 3

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find a pair with a given sum — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

207. Remove duplicates using two pointers — Variation 3

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: remove duplicates using two pointers — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

208. Reverse vowels in a string — Variation 3

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: reverse vowels in a string — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

209. Find maximum sum subarray of size K — Variation 3

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find maximum sum subarray of size k — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

210. Find the longest substring without repeating characters — Variation 3

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find the longest substring without repeating characters — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

211. Find a pair with a given sum — Variation 4

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find a pair with a given sum — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

212. Remove duplicates using two pointers — Variation 4

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: remove duplicates using two pointers — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

213. Reverse vowels in a string — Variation 4

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: reverse vowels in a string — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

214. Find maximum sum subarray of size K — Variation 4

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find maximum sum subarray of size k — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

215. Find the longest substring without repeating characters — Variation 4

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find the longest substring without repeating characters — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

216. Find a pair with a given sum — Variation 5

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find a pair with a given sum — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

217. Remove duplicates using two pointers — Variation 5

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: remove duplicates using two pointers — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

218. Reverse vowels in a string — Variation 5

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: reverse vowels in a string — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

219. Find maximum sum subarray of size K — Variation 5

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find maximum sum subarray of size k — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

220. Find the longest substring without repeating characters — Variation 5

Category: Two Pointers & Sliding Window **Difficulty:** Easy **Tags:** two pointers & sliding window, beginner

Problem Statement: Solve the following task: find the longest substring without repeating characters — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

221. Check whether a number is prime

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

222. Find factorial of N

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

223. Compute GCD of two numbers

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

224. Compute LCM of two numbers

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

225. Reverse an integer

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

226. Check whether a number is an Armstrong number

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is an armstrong number. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

227. Check whether a number is prime — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

228. Find factorial of N — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

229. Compute GCD of two numbers — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

230. Compute LCM of two numbers — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

231. Reverse an integer — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

232. Check whether a number is an Armstrong number — Variation 2

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is an armstrong number — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

233. Check whether a number is prime — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

234. Find factorial of N — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

235. Compute GCD of two numbers — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

236. Compute LCM of two numbers — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

237. Reverse an integer — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

238. Check whether a number is an Armstrong number — Variation 3

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is an armstrong number — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

239. Check whether a number is prime — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

240. Find factorial of N — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

241. Compute GCD of two numbers — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

242. Compute LCM of two numbers — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

243. Reverse an integer — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

244. Check whether a number is an Armstrong number — Variation 4

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is an armstrong number — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

245. Check whether a number is prime — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

246. Find factorial of N — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

247. Compute GCD of two numbers — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

248. Compute LCM of two numbers — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

249. Reverse an integer — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

250. Check whether a number is an Armstrong number — Variation 5

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is an armstrong number — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

251. Check whether a number is prime — Variation 6

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: check whether a number is prime — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

252. Find factorial of N — Variation 6

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: find factorial of n — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

253. Compute GCD of two numbers — Variation 6

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute gcd of two numbers — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

254. Compute LCM of two numbers — Variation 6

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: compute lcm of two numbers — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

255. Reverse an integer — Variation 6

Category: Math & Number Theory **Difficulty:** Easy **Tags:** math & number theory, beginner

Problem Statement: Solve the following task: reverse an integer — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

256. Print numbers from 1 to N recursively

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: print numbers from 1 to n recursively. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

257. Calculate power recursively

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: calculate power recursively. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

258. Find sum of digits recursively

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: find sum of digits recursively. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

259. Generate all subsets of a small array

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all subsets of a small array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

260. Generate all permutations of a string

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all permutations of a string. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

261. Print numbers from 1 to N recursively — Variation 2

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: print numbers from 1 to n recursively — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

262. Calculate power recursively — Variation 2

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: calculate power recursively — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

263. Find sum of digits recursively — Variation 2

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: find sum of digits recursively — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

264. Generate all subsets of a small array — Variation 2

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all subsets of a small array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

265. Generate all permutations of a string — Variation 2

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all permutations of a string — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

266. Print numbers from 1 to N recursively — Variation 3

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: print numbers from 1 to n recursively — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

267. Calculate power recursively — Variation 3

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: calculate power recursively — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

268. Find sum of digits recursively — Variation 3

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: find sum of digits recursively — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

269. Generate all subsets of a small array — Variation 3

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all subsets of a small array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

270. Generate all permutations of a string — Variation 3

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all permutations of a string — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

271. Print numbers from 1 to N recursively — Variation 4

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: print numbers from 1 to n recursively — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

272. Calculate power recursively — Variation 4

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: calculate power recursively — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

273. Find sum of digits recursively — Variation 4

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: find sum of digits recursively — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

274. Generate all subsets of a small array — Variation 4

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all subsets of a small array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

275. Generate all permutations of a string — Variation 4

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all permutations of a string — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

276. Print numbers from 1 to N recursively — Variation 5

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: print numbers from 1 to n recursively — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

277. Calculate power recursively — Variation 5

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: calculate power recursively — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

278. Find sum of digits recursively — Variation 5

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: find sum of digits recursively — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

279. Generate all subsets of a small array — Variation 5

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all subsets of a small array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

280. Generate all permutations of a string — Variation 5

Category: Recursion & Backtracking **Difficulty:** Easy **Tags:** recursion & backtracking, beginner

Problem Statement: Solve the following task: generate all permutations of a string — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

281. Traverse a singly linked list

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

282. Count nodes in a linked list

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

283. Search for a value in a linked list

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

284. Reverse a linked list

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

285. Find the middle node

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

286. Delete the first occurrence of a value

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: delete the first occurrence of a value. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

287. Traverse a singly linked list — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

288. Count nodes in a linked list — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

289. Search for a value in a linked list — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

290. Reverse a linked list — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

291. Find the middle node — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

292. Delete the first occurrence of a value — Variation 2

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: delete the first occurrence of a value — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

293. Traverse a singly linked list — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

294. Count nodes in a linked list — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

295. Search for a value in a linked list — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

296. Reverse a linked list — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

297. Find the middle node — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

298. Delete the first occurrence of a value — Variation 3

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: delete the first occurrence of a value — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

299. Traverse a singly linked list — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

300. Count nodes in a linked list — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

301. Search for a value in a linked list — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

302. Reverse a linked list — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

303. Find the middle node — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

304. Delete the first occurrence of a value — Variation 4

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: delete the first occurrence of a value — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

305. Traverse a singly linked list — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

306. Count nodes in a linked list — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

307. Search for a value in a linked list — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

308. Reverse a linked list — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

309. Find the middle node — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

310. Delete the first occurrence of a value — Variation 5

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: delete the first occurrence of a value — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

311. Traverse a singly linked list — Variation 6

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: traverse a singly linked list — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

312. Count nodes in a linked list — Variation 6

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: count nodes in a linked list — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

313. Search for a value in a linked list — Variation 6

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: search for a value in a linked list — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

314. Reverse a linked list — Variation 6

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: reverse a linked list — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

315. Find the middle node — Variation 6

Category: Linked Lists **Difficulty:** Easy **Tags:** linked lists, beginner

Problem Statement: Solve the following task: find the middle node — variation 6. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

316. Implement a stack using an array

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: implement a stack using an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

317. Check balanced parentheses

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: check balanced parentheses. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

318. Reverse a string using a stack

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: reverse a string using a stack. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

319. Find the next greater element

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: find the next greater element. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

320. Evaluate a postfix expression

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: evaluate a postfix expression. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

321. Implement a stack using an array — Variation 2

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: implement a stack using an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

322. Check balanced parentheses — Variation 2

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: check balanced parentheses — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

323. Reverse a string using a stack — Variation 2

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: reverse a string using a stack — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

324. Find the next greater element — Variation 2

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: find the next greater element — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

325. Evaluate a postfix expression — Variation 2

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: evaluate a postfix expression — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

326. Implement a stack using an array — Variation 3

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: implement a stack using an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

327. Check balanced parentheses — Variation 3

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: check balanced parentheses — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

328. Reverse a string using a stack — Variation 3

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: reverse a string using a stack — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

329. Find the next greater element — Variation 3

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: find the next greater element — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

330. Evaluate a postfix expression — Variation 3

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: evaluate a postfix expression — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

331. Implement a stack using an array — Variation 4

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: implement a stack using an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

332. Check balanced parentheses — Variation 4

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: check balanced parentheses — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

333. Reverse a string using a stack — Variation 4

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: reverse a string using a stack — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

334. Find the next greater element — Variation 4

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: find the next greater element — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

335. Evaluate a postfix expression — Variation 4

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: evaluate a postfix expression — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

336. Implement a stack using an array — Variation 5

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: implement a stack using an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

337. Check balanced parentheses — Variation 5

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: check balanced parentheses — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

338. Reverse a string using a stack — Variation 5

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: reverse a string using a stack — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

339. Find the next greater element — Variation 5

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: find the next greater element — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

340. Evaluate a postfix expression — Variation 5

Category: Stacks **Difficulty:** Easy **Tags:** stacks, beginner

Problem Statement: Solve the following task: evaluate a postfix expression — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

341. Implement a queue using an array

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & deques, beginner

Problem Statement: Solve the following task: implement a queue using an array. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

342. Implement a circular queue

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & deques, beginner

Problem Statement: Solve the following task: implement a circular queue. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

343. Generate binary numbers from 1 to N

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: generate binary numbers from 1 to n. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

344. Reverse the first K queue elements

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: reverse the first k queue elements. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

345. Implement a queue using an array — Variation 2

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a queue using an array — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

346. Implement a circular queue — Variation 2

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a circular queue — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

347. Generate binary numbers from 1 to N — Variation 2

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: generate binary numbers from 1 to n — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

348. Reverse the first K queue elements — Variation 2

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: reverse the first k queue elements — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

349. Implement a queue using an array — Variation 3

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a queue using an array — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

350. Implement a circular queue — Variation 3

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a circular queue — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

351. Generate binary numbers from 1 to N — Variation 3

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: generate binary numbers from 1 to n — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

352. Reverse the first K queue elements — Variation 3

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: reverse the first k queue elements — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

353. Implement a queue using an array — Variation 4

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a queue using an array — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

354. Implement a circular queue — Variation 4

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a circular queue — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

355. Generate binary numbers from 1 to N — Variation 4

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: generate binary numbers from 1 to n — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

356. Reverse the first K queue elements — Variation 4

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: reverse the first k queue elements — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

357. Implement a queue using an array — Variation 5

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a queue using an array — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

358. Implement a circular queue — Variation 5

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: implement a circular queue — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

359. Generate binary numbers from 1 to N — Variation 5

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: generate binary numbers from 1 to n — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

360. Reverse the first K queue elements — Variation 5

Category: Queues & Deques **Difficulty:** Easy **Tags:** queues & dequeues, beginner

Problem Statement: Solve the following task: reverse the first k queue elements — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

361. Count nodes in a binary tree

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count nodes in a binary tree. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

362. Find tree height

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find tree height. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

363. Preorder traversal

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: preorder traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

364. Inorder traversal

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: inorder traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

365. Postorder traversal

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: postorder traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

366. Level order traversal

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: level order traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

367. Count leaf nodes

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count leaf nodes. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

368. Find maximum value in a binary tree

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find maximum value in a binary tree. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

369. Count nodes in a binary tree — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count nodes in a binary tree — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

370. Find tree height — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find tree height — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

371. Preorder traversal — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: preorder traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

372. Inorder traversal — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: inorder traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

373. Postorder traversal — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: postorder traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

374. Level order traversal — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: level order traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

375. Count leaf nodes — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count leaf nodes — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

376. Find maximum value in a binary tree — Variation 2

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find maximum value in a binary tree — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

377. Count nodes in a binary tree — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count nodes in a binary tree — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

378. Find tree height — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find tree height — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

379. Preorder traversal — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: preorder traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

380. Inorder traversal — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: inorder traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

381. Postorder traversal — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: postorder traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

382. Level order traversal — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: level order traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

383. Count leaf nodes — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count leaf nodes — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

384. Find maximum value in a binary tree — Variation 3

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find maximum value in a binary tree — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

385. Count nodes in a binary tree — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count nodes in a binary tree — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

386. Find tree height — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find tree height — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

387. Preorder traversal — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: preorder traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

388. Inorder traversal — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: inorder traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

389. Postorder traversal — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: postorder traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

390. Level order traversal — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: level order traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

391. Count leaf nodes — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count leaf nodes — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

392. Find maximum value in a binary tree — Variation 4

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find maximum value in a binary tree — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

393. Count nodes in a binary tree — Variation 5

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: count nodes in a binary tree — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

394. Find tree height — Variation 5

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: find tree height — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

395. Preorder traversal — Variation 5

Category: Binary Trees **Difficulty:** Easy **Tags:** binary trees, beginner

Problem Statement: Solve the following task: preorder traversal — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

396. Search in a BST

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: search in a bst. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

397. Insert into a BST

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: insert into a bst. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

398. Find minimum in a BST

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find minimum in a bst. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

399. Find maximum in a BST

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find maximum in a bst. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

400. Validate a BST

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: validate a bst. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

401. Search in a BST — Variation 2

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: search in a bst — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

402. Insert into a BST — Variation 2

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: insert into a bst — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

403. Find minimum in a BST — Variation 2

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find minimum in a bst — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

404. Find maximum in a BST — Variation 2

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find maximum in a bst — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

405. Validate a BST — Variation 2

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: validate a bst — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

406. Search in a BST — Variation 3

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: search in a bst — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

407. Insert into a BST — Variation 3

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: insert into a bst — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

408. Find minimum in a BST — Variation 3

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find minimum in a bst — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

409. Find maximum in a BST — Variation 3

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: find maximum in a bst — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

410. Validate a BST — Variation 3

Category: Binary Search Trees **Difficulty:** Easy **Tags:** binary search trees, beginner

Problem Statement: Solve the following task: validate a bst — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

411. Find the Kth smallest element

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth smallest element. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

412. Find the Kth largest element

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth largest element. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

413. Build a min heap

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: build a min heap. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

414. Merge K sorted small arrays

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: merge k sorted small arrays. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

415. Find the Kth smallest element — Variation 2

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth smallest element — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

416. Find the Kth largest element — Variation 2

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth largest element — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

417. Build a min heap — Variation 2

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: build a min heap — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

418. Merge K sorted small arrays — Variation 2

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: merge k sorted small arrays — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

419. Find the Kth smallest element — Variation 3

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth smallest element — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

420. Find the Kth largest element — Variation 3

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth largest element — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

421. Build a min heap — Variation 3

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: build a min heap — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

422. Merge K sorted small arrays — Variation 3

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: merge k sorted small arrays — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

423. Find the Kth smallest element — Variation 4

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth smallest element — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

424. Find the Kth largest element — Variation 4

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: find the kth largest element — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

425. Build a min heap — Variation 4

Category: Heaps & Priority Queues **Difficulty:** Easy **Tags:** heaps & priority queues, beginner

Problem Statement: Solve the following task: build a min heap — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

426. Breadth-first traversal

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: breadth-first traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

427. Depth-first traversal

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: depth-first traversal. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

428. Count connected components

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: count connected components. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

429. Check whether a path exists

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: check whether a path exists. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

430. Find degree of a vertex

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: find degree of a vertex. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

431. Breadth-first traversal — Variation 2

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: breadth-first traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

432. Depth-first traversal — Variation 2

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: depth-first traversal — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

433. Count connected components — Variation 2

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: count connected components — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

434. Check whether a path exists — Variation 2

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: check whether a path exists — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

435. Find degree of a vertex — Variation 2

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: find degree of a vertex — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

436. Breadth-first traversal — Variation 3

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: breadth-first traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

437. Depth-first traversal — Variation 3

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: depth-first traversal — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

438. Count connected components — Variation 3

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: count connected components — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

439. Check whether a path exists — Variation 3

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: check whether a path exists — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

440. Find degree of a vertex — Variation 3

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: find degree of a vertex — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

441. Breadth-first traversal — Variation 4

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: breadth-first traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

442. Depth-first traversal — Variation 4

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: depth-first traversal — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

443. Count connected components — Variation 4

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: count connected components — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

444. Check whether a path exists — Variation 4

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: check whether a path exists — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

445. Find degree of a vertex — Variation 4

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: find degree of a vertex — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

446. Breadth-first traversal — Variation 5

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: breadth-first traversal — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

447. Depth-first traversal — Variation 5

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: depth-first traversal — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

448. Count connected components — Variation 5

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: count connected components — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

449. Check whether a path exists — Variation 5

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: check whether a path exists — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

450. Find degree of a vertex — Variation 5

Category: Graphs **Difficulty:** Easy **Tags:** graphs, beginner

Problem Statement: Solve the following task: find degree of a vertex — variation 5. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

451. Activity selection for small intervals

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: activity selection for small intervals. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

452. Assign cookies to children

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: assign cookies to children. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

453. Find minimum coins for canonical denominations

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: find minimum coins for canonical denominations. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

454. Schedule jobs by deadline

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: schedule jobs by deadline. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

455. Activity selection for small intervals — Variation 2

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: activity selection for small intervals — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

456. Assign cookies to children — Variation 2

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: assign cookies to children — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

457. Find minimum coins for canonical denominations — Variation 2

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: find minimum coins for canonical denominations — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

458. Schedule jobs by deadline — Variation 2

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: schedule jobs by deadline — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

459. Activity selection for small intervals — Variation 3

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: activity selection for small intervals — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

460. Assign cookies to children — Variation 3

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: assign cookies to children — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

461. Find minimum coins for canonical denominations — Variation 3

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: find minimum coins for canonical denominations — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

462. Schedule jobs by deadline — Variation 3

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: schedule jobs by deadline — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

463. Activity selection for small intervals — Variation 4

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: activity selection for small intervals — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

464. Assign cookies to children — Variation 4

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: assign cookies to children — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

465. Find minimum coins for canonical denominations — Variation 4

Category: Greedy **Difficulty:** Easy **Tags:** greedy, beginner

Problem Statement: Solve the following task: find minimum coins for canonical denominations — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

466. Fibonacci using memoization

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: fibonacci using memoization. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

467. Climbing stairs

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: climbing stairs. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

468. Minimum cost to reach the end

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: minimum cost to reach the end. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

469. House robber with non-negative values

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: house robber with non-negative values. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

470. Count ways to make a small sum

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: count ways to make a small sum. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

471. Fibonacci using memoization — Variation 2

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: fibonacci using memoization — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

472. Climbing stairs — Variation 2

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: climbing stairs — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

473. Minimum cost to reach the end — Variation 2

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: minimum cost to reach the end — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

474. House robber with non-negative values — Variation 2

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: house robber with non-negative values — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

475. Count ways to make a small sum — Variation 2

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: count ways to make a small sum — variation 2. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

476. Fibonacci using memoization — Variation 3

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: fibonacci using memoization — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

477. Climbing stairs — Variation 3

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: climbing stairs — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

478. Minimum cost to reach the end — Variation 3

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: minimum cost to reach the end — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

479. House robber with non-negative values — Variation 3

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: house robber with non-negative values — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

480. Count ways to make a small sum — Variation 3

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: count ways to make a small sum — variation 3. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

481. Fibonacci using memoization — Variation 4

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: fibonacci using memoization — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

482. Climbing stairs — Variation 4

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: climbing stairs — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

483. Minimum cost to reach the end — Variation 4

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: minimum cost to reach the end — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

484. House robber with non-negative values — Variation 4

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: house robber with non-negative values — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.

485. Count ways to make a small sum — Variation 4

Category: Dynamic Programming **Difficulty:** Easy **Tags:** dynamic programming, beginner

Problem Statement: Solve the following task: count ways to make a small sum — variation 4. Read the input, perform the required operation, and print the result. Your solution should handle all valid inputs within the stated constraints.

Constraints: $1 \leq N \leq 10^5$ unless otherwise stated; values fit in a signed 32-bit integer; input contains valid data.

Input Format: The first line contains N. The next line contains N space-separated values, or the string/parameters required by the problem.

Output Format: Print the required answer on one line. Do not print extra explanatory text.

Example: Input: 5 | 1 2 3 4 5 Output: 5 (illustrative; adapt according to the exact task).

Visible Test Cases: (1) Smallest valid input. (2) Typical mixed input. (3) Already processed or uniform input. (4) Boundary-sized values.

Hidden Test Cases: Include minimum N, maximum N, duplicates, negative values where valid, all-equal values, sorted and reverse-sorted data, empty-result scenarios where valid, and overflow-sensitive sums/products.

Judge Notes: Compare normalized output exactly; reject extra text. Add category-specific edge cases and randomized tests before publishing.